

|          |                                                                                                                                                                                   |           |
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|          |                                                                                                                                                                                   |           |
| 1(a)(i)  | force / area (normal to the force)                                                                                                                                                | <b>B1</b> |
| 1(a)(ii) | $(p = F / A \text{ so units are}) \text{ kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{ s}^{-2}$                                                                             | <b>A1</b> |
| 1(b)     | unit of $R$ : m <b>and</b> unit of $t$ : s <b>and</b> unit of $L$ : m                                                                                                             | <b>C1</b> |
|          | unit of $\rho$ : $\text{kg m}^{-3}$<br><b>or</b><br>$\rho = m / V$                                                                                                                | <b>C1</b> |
|          | base units of $k$ : $(\text{kg m}^{-1} \text{ s}^{-2} \times \text{m}^4 \times \text{kg m}^{-3} \times \text{s}) / (\text{kg} \times \text{m}) = \text{kg m}^{-1} \text{ s}^{-1}$ | <b>A1</b> |
| 1(c)     | $R$ contributes $4 \times 2\%$ <b>or</b> $8\%$ (and $L$ contributes $2\%$ ) so $R$ contributes more (to the percentage uncertainty in $k$ )                                       | <b>B1</b> |

| Question | Answer | Marks |
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